

ASSESSMENT 1 PRACTICE PROBLEMS

DATA 20195 / 30195
MOON DUCHIN, SPRING 2026

From notebooks.

- (1) Suppose a dataframe called `df` has one column that contains the number of Clinton votes and another that contains the number of Trump votes. How do you add a column to your dataframe that gives the Clinton share of the vote? Either write code to do so or give a clear explanation of the database operations involved.
- (2) Here's a code block:

```
import pandas as pd
data = { "A": [5, 15, 25, 35, 45, 55, 65], "B": [10, 20, 30, 40, 50, 60, 70],
        "C": [2, 4, 6, 8, 10, 12, 14] }
df = pd.DataFrame(data)
```

Questions

 - Is B a row or column of this dataframe?
 - what does `df.head(2)` return?
 - write an expression for the sum $10 + 20 + \dots + 70$ using this dataframe.- (3) The Census uses a 15-digit numerical identifier called a GEOID for census blocks. Explain (in English) how to find the state and county, and either write code to do so or give a clear explanation of the database operations that are involved.

From readings.

True/False and explain. *Your answer should show some knowledge about the topics discussed.*

- (4) Today, several countries are still fighting over wedges of Antarctica.
- (5) In Brazil, if you are racially mixed, you are considered Black under the so-called “One Drop Rule.”

Short answer.

- (6) Melissa Nobles tells us that in the Soviet Union in the 1930s, “Senior statisticians who had supervised the census were arrested and shot.” Why? (And even if you don’t remember this specific example, what could be so scandalous in a census?)
- (7) James Scott talks about high modernism as “a strong, one might even say muscle-bound, version of the self-confidence about scientific and technical progress.” Is this automatically dangerous — and if not, what is needed for it to become dangerous?

From lecture.

True/False and explain. *Your answer should show some knowledge about the topics discussed.*

- (8) The South African apartheid government did not need a bureaucracy for race classification because it was defined and enforced purely socially.
- (9) Raster data is essentially outdated and is not used in modern computer-based geography.
- (10) Not every state has census blocks, and for the ones that do, they nearly always have between 100 and 500 people.

Short answer.

- (11) In Ian Hacking’s “making up”/“looping” framework, were there autistic people in the ancient world?
- (12) In what sense does zoning resemble confiscation, or taking land away from landowners?

How I figure time: budget 5 minutes for each of the numbered questions. This full set of questions would be budgeted at 60 minutes. Your assessment (mini-test) next week will be shorter—two code and two reading/writing questions—budgeted for 20 minutes, but you can feel free to take 30 minutes.